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Bachelor Thesis

Data Visualization for Augmented Reality Head-Up Displays in Personal Protective Equipment

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1 Introduction

While working in personal protection equipment, it is essential to be able to parse critical information, such as oxygen levels, at a glance. Chemical protective suits in particular reduce peripheral vision and mobility [Murray et al., 2011], making it challenging to operate radio devices and other tools.

A possible solution might be a head-up display (HUD), since it could integrate information into the field of view. Numerous studies confirm an increase in task performance when using a HUD [Kim et al., 2019].

This thesis aims to analyze requirements and current technological issues, as well as propose potential user interface designs for HUDs independent of hardware implementation.

I recommend testing yourself if the design works in the peripheral vision.

2 Chemical Protective Clothing and Its Applications

Scenarios where workers are exposed to hazardous materials require chemical protective clothing (CPC). Protection levels range from limited spray protection (Type 6) to fully encapsulating, gas-tight suits (Type 1) with self-contained breathing apparatus (SCBA) ["DGUV Information 212-019"].

Figure 1: Type 1 chemical protection suit 4005/NEO manufactured by isotemp[1]

Confined spaces are defined as areas in which, through their design or site conditions, "inadequate air replacement rates, or the substances [...] found within them or introduced into them, particular hazards exist or may arise that substantially exceed the normal prevailing hazard potential of workplaces" ["DGUV Regel 113-004"]. Tasks that require entering confined spaces are typically cleaning, inspection and/or maintenance of, for example, tanks in chemical plants, rail cars and highway tank trailers and even sewage containers. Traces of the contents may remain, even if their containers are prepared correctly. These residues might—despite being initially inactive—become dangerous through work procedures (e.g. welding), agitation or chemical reactions ["DGUV Regel 113-004"], hence requiring the highest levels of protection. Other use cases are emergencies, such as

[1]Vollschutzanzug 4005/NEO · Chemie · isotemp

confined space rescue, removing chemical spills or other accidents involving unknown, possibly hazardous substances [“OSHA Technical Manual Section VIII”].

These tasks are highly demanding—both physically and mentally—particularly because of the equipment. Since Type 1 suits are not gas-permeable, they compromise the body’s ability to regulate temperature, leading to exhaustion and heat fatigue even faster [Holmér, 2006]. The exertion is exacerbated by heavier equipment and heat stress, ultimately reducing work capacity [Smolander et al., 1984]. Maximum working time in a full-body suit with SCBA is therefore limited and might decrease even more depending on air and exhaustion levels.

Further, the material and shape of the suit muffle hearing and limit mobility and the field of view to such an extent that even trained personnel might need double the time to complete tasks when wearing CPC [Murray et al., 2011].

3 Augmented Reality

Augmented reality (AR) is a subcategory of extended reality (XR), next to mixed reality (MR) and virtual reality (VR). In AR virtual objects are "superimposed upon or composited with the real world" [Azuma, 1997], in opposition to virtual reality which—by replacing the real environment entirely—creates the deepest sense of immersion. Mixed reality combines features of both, the main distinction being the ability to interact with virtual objects in the real world [Hari Chandana et al., 2023].

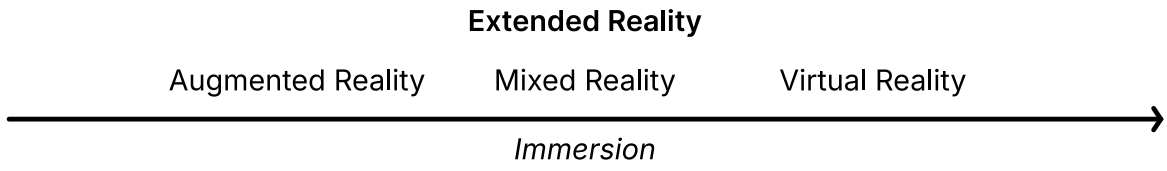

Figure 2: The extended reality spectrum, encompassing augmented, mixed and virtual reality.

Azuma (1997) defines three requirements for applications to be classified as AR. They combine real and virtual content, they are interactive in real-time, and the virtual objects are registered in 3D space. With the technological advancements since the 2000s these requirements are understood more as guidelines than rules.

A head-up display overlays information onto reality in the user's line of sight to avoid the need to look away from their primary task to acquire needed information. Originally used in military aircraft, today HUDs are built into consumer vehicles as well. Apart from these integrated HUDs, content can be presented on a head-worn display. This affixes the display to the head, typically as a helmet or glasses, enabling mobile usage. Contrary to the above definition by Azuma (1997), the content displayed on a HUD is two-dimensional and head-stabilized, rather than anchored spatially, i.e. world-stabilized (see figure 3). Nevertheless, some authors, such as Lu et al. (2020), categorize HUDs as AR.

A head-worn display can cover one (monocular) or both eyes. Depending on the technology, both eyes receive the same image (biocular) or each will be shown a different image (binocular).

Field of View. Syberfeldt et al. (2017) sets the threshold for an acceptable field of view at a minimum of 30° for industrial applications. Microsoft's HoloLens 2 features a 43° by 29° field of view, while a few other systems reach around 40°.[2]

[2] Compare all AR Headsets & Glasses - VRcompare

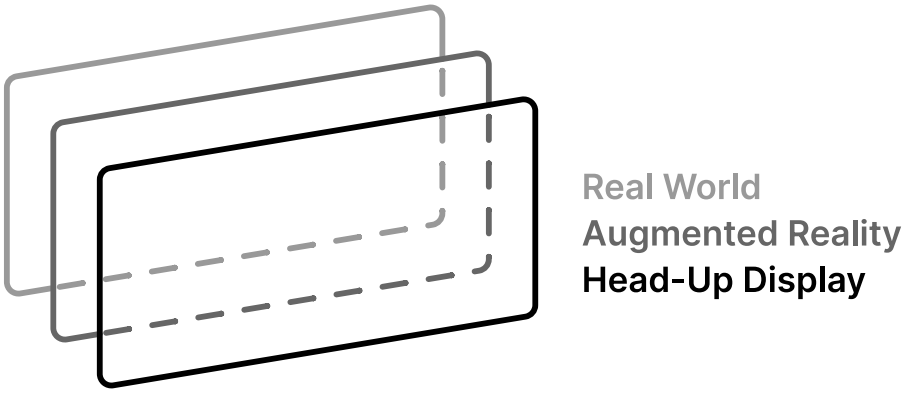

Figure 3: Visualization layers of head-worn displays: reality, Augmented Reality (3D, world-stabilized) and User Interface (2D, head-stabilized).

Screen Shake. As with all head-worn devices, HMDs are susceptible to shaking when moving, which causes legibility issues [Janaka, Zhao, and Chan, 2023]. Especially thin, horizontal lines tend to disappear optically due to the up- and down movement.

3.1 Optical See-Through Displays

There are two approaches for combining virtual content with reality. With video see-through, information will be superimposed on a video stream filmed by a camera on the headset. These headsets fully occlude sight, the stream replaces the outside view. An optical see-through system allows the user to directly view reality.

When a video see-through system fails, it will be impossible to see. Failing equipment must not impair vision, especially since it cannot be taken off in a fully encapsulated suit.

3.1.1 Ambient Light and Contrast Issues

Regardless of the optic technology used, images are created by combining light, which means the display has to compete with ambient light for visibility. This results in a significant loss of contrast, especially outdoors. Kerr et al. (2011), for example, report that participants repeatedly used their hands to shield the display from external light to be able to read the its contents.

Many manufacturers add shaded lenses to improve the contrast somewhat [Itoh

et al., 2021]. Unfortunately, current AR technology is not capable of rendering the contrast levels necessary for outdoor use [Erickson et al., 2020]. Technical solutions in development include OSTs that can partially occlude areas of the display to block light. Still very experimental [Davari et al., 2020].

On physical screens, the color we see corresponds to the actual color of the display material. Since OSTs do not have a physical screen, "black" is transparent [Gabbard et al., 2013]. This also means that darker colors are not available for design on OSTs, as they are mostly translucent. Many current OSTs, such as [MODEL], feature tinted lenses to slightly improve contrast [Itoh et al., 2021]. Unfortunately, current AR technology is unable to display the contrast levels required for outdoor use [Erickson et al., 2020]. Technical solutions currently under development include OSTs that can partially cover areas of the display to block light. However, this is still very experimental [Davari et al., 2020].

3.1.2 Color Blending

In the additive color model, the combination of all wavelengths of light produce white, while black is created by the absence of light. When physical screens are turned off, the actual color of the display material is visible. Since OSTs do not have a physical screen, "black" is transparent and darker colors mostly translucent [Gabbard et al., 2013]. Additionally, the level of transparency affects the extent to which the background distorts the color [Itoh et al., 2021].

3.2 Methods of Interaction

Due to not being constrained to two dimensions, XR applications can explore new types of interaction, such as gesture recognition, speech recognition and gaze tracking. However, industrial use cases require hands-free interaction methods. Additionally, noise levels in these environments limit possible interaction to gaze tracking. Even with these constraints, Hari Chandana et al. (2023) emphasize the importance of interactions being intuitive and feeling natural.

Gaze-based interaction is enabled by eye tracking technology. In addition to tracking eye position, it analyzes where the user is looking and how long.

4 The Human Visual System

4.1 The Human Visual System

A simplified model estimates the field of view to be approximately 200° to 220° horizontally and around 125° vertically, with the vertical field of view extending slightly further downward [Loschky et al., 2017]. The central vision covers about 5° around the point of fixation, where acuity is highest [Loschky et al., 2017, Rosenholtz, 2016]. The remaining field is the periphery, with boundaries at 30° and 60°, dividing it into near, mid and far peripheral vision.

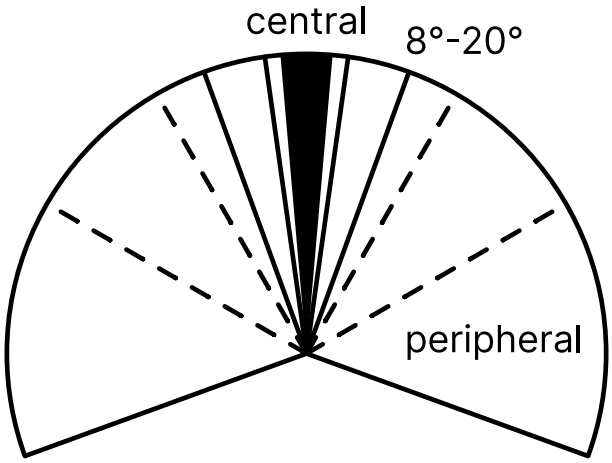

Figure 4: Simplified illustration of the horizontal field of view.

Despite visual acuity and other functions deteriorating in the periphery, the area in front of us continues to appear in high resolution, with high contrast and color saturation [Loschky et al., 2017]. According to Rosenholtz (2016) the decline in performance of the peripheral vision is smaller than commonly assumed. A scene does not require many details to still allow object recognition and orientation, which is why image thumbnails remain recognizable despite their small size.

Rosenholtz (2016) suggests that crowding is the biggest issue in parsing information located in the periphery. The effect is more pronounced if the target and its neighboring elements are similar (e.g., same color and shape) [Pelli and Tillman, 2008]. The farther elements are from the center of vision, the greater the distance between them must be.

In addition, peripheral vision is particularly well suited for detecting motion, provides better vision in low-light conditions [Johnson, 2014] and contributes to functions like orientation, navigation, balance, and hand-eye coordination. A restric-

tion of the peripheral visual field severely impairs the performance of these functions, the effect increasing with the amount of vision obstructed [Alfano and Michel, 1990].

Color sensitivity decreases toward the periphery, red-green moreso than blue-yellow colors and sensitivity to luminance. Hansen et al. (2009) found that color perception is possible even at an eccentricity of 50°, provided the stimulus is sufficiently large. Most modern OST-HMDs feature a 40° field of view, meaning the entire available color spectrum can be utilized.

4.2 Visual Attention

The retina transmits too much information to be processed all at once. This bottle neck is resolved by visual attention—by limiting processing to focus on just a fragment of the current visual stimuli [Anderson et al., 2005, Rosenholtz, 2016]. This process is guided by peripheral vision, which subconsciously preselects salient features of the environment [**<empty citation>**]. Visual salience describes features that distinguish a target from its surrounding objects [Johnson, 2014].

5 Attention Management

Bottom-up attention occurs subconsciously, triggered by outside stimuli. In the top-down process, the brain directs the eyes toward a consciously selected object. Interestingly, top-down attention lessens the effect of distracting stimuli when concentrating on a target [Connor et al., 2004].

When multiple stimuli compete for attention, even salient targets might be ignored if they are irrelevant to the current goal. This is referred to as inattentional blindness [Jensen et al., 2011]. Cater et al. (2002) leverages this effect to only render parts of frames in high-quality around the current fixation point.

5.1 Situation Awareness

The concept of situation awareness provides context for the visualizations.

Making good judgements becomes increasingly difficult when performing demanding procedures while trying to maintain awareness of the surrounding area. The risks associated with these activities prohibit human error, further increasing stress levels. Endsley (1995) highlights the role of situation awareness as a foundation for good decision-making in dynamic situations. She outlines three levels of SA, namely perception, comprehension and projection, of which perception is the most significant for UI design. This stage requires collecting the information relevant to the situation. In level 2 this information will be analyzed to form "a holistic picture of the environment, comprehending the significance of objects and events." Finally, likely outcomes can be determined by understanding the dynamics of the situation.

It is evident that situation awareness is therefore dependent on the processes of attention described in section 4. Subsequently, this illustrates the necessity of a thorough understanding of the pieces required for achieving good situation awareness to design a well-functioning HUD. The amount of information that can be rapidly and accurately obtained is directly influenced by how it is presented. In addition to prioritizing and organizing that information around task objectives, Endsley (1995) mentions the advantage of systems providing processed information rather than the data it is based on (e.g. displaying remaining time based on current oxygen amount and usage).

Special attention should be paid to reserve highly salient features to critical events that truly necessitate changing the course of action. Further, it is vital to maintain a balance between showing all relevant information, while simultaneously avoiding visual overload. While looking for inaccessible information, other important content may be overlooked. Conversely, "filtering of extraneous information (not related to SA needs) and reduction of data (by processing and integrating low-level data to arrive at SA requirements)" is necessary to prevent clutter. This reiterates the importance of accurately evaluating and prioritizing task-relevant information.

Considering these recommendations supports evaluating the interaction of all components, since emphasizing some elements inevitably comes at the expense of others. Assessing the design concept by the amount of situation awareness it provides serves as the last step of ensuring the effectiveness of the system.

5.2 Peripheral Displays and Glanceability

A peripheral display keeps information related to a primary task available. On the other hand, so-called notification displays are meant to prompt action, actually interrupting the user. A peripheral display offers the choice of whether or not to continue the current action [Matthews et al., 2005].

Glanceable information means it can be comprehended quickly and easily once it is paid attention to [Matthews et al., 2006].

Glanceability is directly impacted by the displayed amount of items, their salience, etc.—i.e. the attentional limits described in section 4.

Interpreting a display of one piece of information will be much easier than interpreting a display of many bits of information. Also affecting glaceability are design elements such as color, shape, position, size, and contrast, which are used to represent information in a variety of ways.

Instead of showing the interface at all times, its elements could be positioned outside of the HMD's field of view, accessible by turning the head in that direction. This is the core idea of "Glanceable AR".

Lu et al. (2020) proposes two concepts to achieve this, comparing it with a typical 2D HUD.

Head-Glance. Interface components are placed just outside the display's field

of view and are accessed turning the head. This approach requires an additional sensor for the position of the body, so the head movement can be tracked in relation to the body.

Gaze-Summon. Here, anchors are placed near the position of where the elements would be placed. If this point is fixated for a certain amount of time, the information will appear in its stead.

This further ties into context-aware interfaces, which display information based on current task or cognitive load Rhodes, 1998. The risk of false positives is unjustifiable in safety-critical environments.

6 Data Visualization

The aim of data visualization is to make large datasets understandable. Aside from design fundamentals, most specialized resources discuss visualization types intended for interrelated data. Interpreting such complex information requires time and cognitive capacity unavailable in scenarios such as those mentioned in section 2. In these situations, a clear picture of the current status must be available at all times, which only allows for individual data values and limits the number of possible visualization types.

Ferdio, an agency specialized on infographics and data visualization, has created a library of visualization types[3]. In this library, out of a 166 options, nine are for single data.

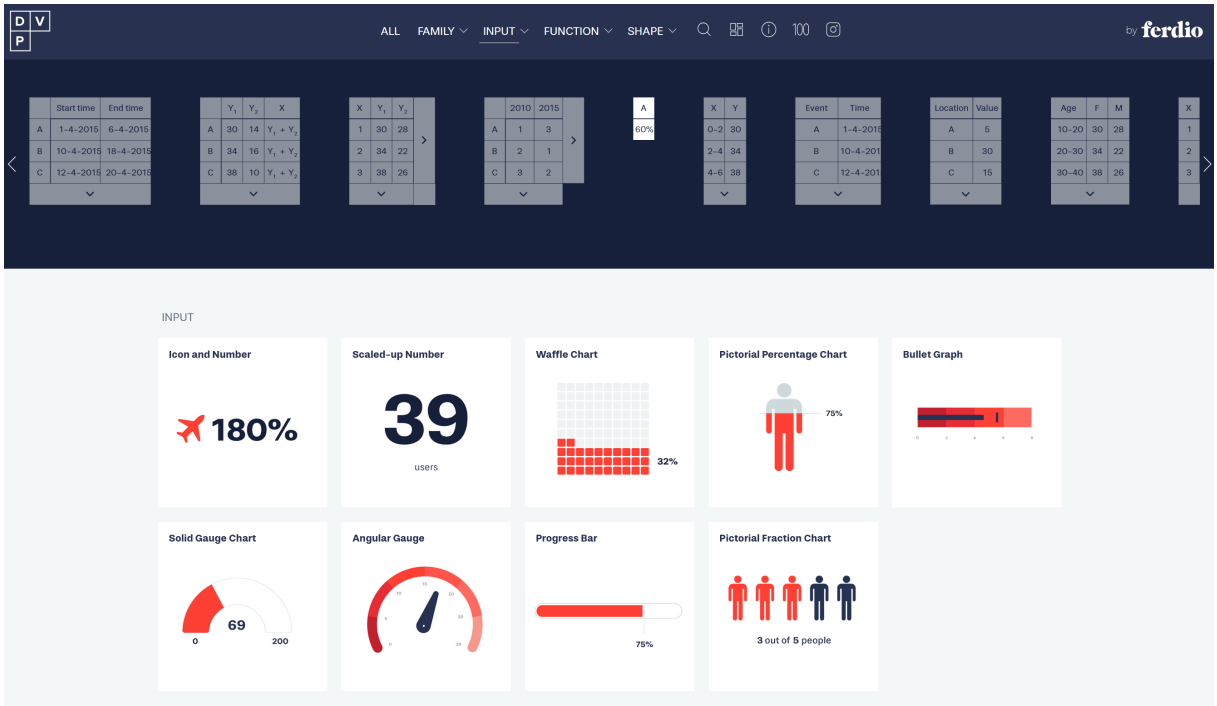

Figure 5: The options for single data (highlighted in the menu bar) in Ferdio's data visualization library.

Wilke (2019) lists position, shape, size and color as basic elements for visualization. Position describes the placement of data, related to other points or axes of a diagram. For color hue and value should be distinguished, since they are suited for different kinds of data. Hues tend to work better for distinctive categories. Value for continuous data GRADIENTS, the eye is much more sensitive to differences in lightness (see section 4). Hues can also be used for gradients (e.g. rainbow), but

[3]Data Viz Project | Collection of data visualizations to get inspired and find the right type

require much more intention to work.

Each of these elements can be used to encode information. Which element is used for which information influences the type of visualization. Importantly, this mapping must be unique, else it becomes ambiguous, muddling the meaning [Wilke, 2019].

6.1 Visualizations for Single Value Data

Icons and Scaled-up Numbers. highlight a single value Often used when the number does not need a context or a comparison icons serve as description or label

Differential Indicators.

These are a type of status indicator, and meant to visualize changes of information.

Solid and Angular Gauges. solid necessarily show specific data/numbers, more for abstract status evaluation Angular Gauge uses a radial scale for specific data points (e.g. temperature) dial over a radial scale with defined limits

According to Xia et al. (2025), bar charts outperform radial visualization.

Status Bar. A progress bar is a graphical control element used to visualize the progression of an extended operation, such as a download, file transfer, or installation. Sometimes, the graphic is accompanied by a textual representation of the progress in a percent or quantitative format.

7 Design Guidelines

The following sections summarize the scenario requirements, technical limitations and human factors.

Ideally, the HUD shows as much information as possible without limiting situation awareness, causing fatigue or sensory overload. The displayed information should be comprehensible in as short a time as possible.

In an emergency, the system must be able to get the user's attention, without hindering decision-making or interrupting the primary task completely.

7.1 Color and Contrast

As mentioned in section 3, the use of color is mainly limited by current optical see-through technology. Gabbard et al. (2022) advises to prefer colors with high source luminance, since these are less subject to color blending. Further, if a color usually signifies a specific meaning and cannot be guaranteed to be perceived correctly, "designers should instead consider other color(s) that exhibit higher luminance but are still meaningfully interpretable". They specifically mention red as such an example.

The library chroma.js[4] has a funtion to sort colors by luminance in the RGB color space, based on the Web Content Accessibility Guidelines 2.0.[5] Sorting the base colors red (#FF0000), green (#00FF00), blue (#0000FF) and their intermediaries cyan (#00FFFF), magenta (#FF00FF), and yellow (#FFFF00) puts them into the following order (brightest to darkest): yellow, cyan, green, magenta, red, and blue.

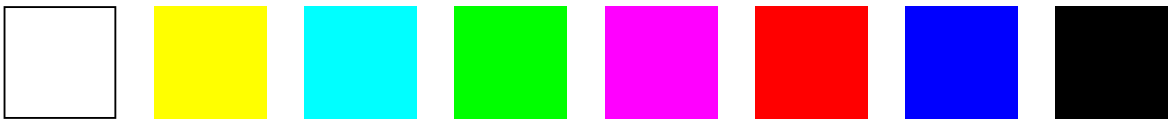

Figure 6: Base colors sorted by luminance.

Limiting the UI to a single color can ensure that the design works effectively in every respect (e.g., by using motion to draw attention instead of color). This also

[4] chroma.js api docs!
[5] Web Content Accessibility Guidelines (WCAG) 2.0

offers economic benefits, as in scenarios where devices are likely to need replacement due to damage or contamination, acquisition costs should be kept as low as possible. One way to ensure affordability is the usage of a monochrome display. This further avoids impairing users with color blindness.

7.2 Typography

Aside from screen shake, reading performance is impacted by lighting and background texture, reiterating the importance of proper contrast.

To find requirements for font size, Renkewitz et al. (2008) tested to what extent background font colors affect text recognition. They set a benchmark of two seconds as threshold for fast recognition, which resulted in recommended font sizes from 23 to 42 pt, depending on color and background. However, these numbers might have to be adjusted for an OST-HMD, since they used a video see-through system, and red and blue as colors, both colors with lower luminance.

As explained in 3, thinner lines disappear when the HMD moves. Matsuura et al. (2019) confirm that heavier font weights improve readability. Naturally, the same requirements exist for stroke width.

For example, the typeface "Inter" fits the requirements well. It has a uniform thickness of strokes, a square letter form and good spacing between its lines.

7.3 Iconography

Janaka, Zhao, and Sapkota (2023) tested if, and under which conditions icons perform better than text. Icon-augmented notifications enabled significantly higher primary task performance than text only and reduced the level of interruption while maintaining reaction and comprehension. They also found icons are more comprehensible when task complexity increased or demanded more attention. The notification were "easier to perceive in suboptimal conditions", including outdoors when walking.

Naturally, this requires the icons to be well-designed. To that end, Lin et al. (2016) tested versions of icons against each other. The differences tested were polarity, plane- and shape-based, and outline or not. Since line-based icons dissect

the space more often, adding visual complexity, which lowers recognition performance. Additionally, white on black icons performed best.

Another reason to prefer shape-based icons is the requirement of thicker line widths which might impact an icon's readability. Simpler icons with varied silhouettes offset this.

Figure 7: Line-based (left) and plane-based (right) icons.

7.4 Polarity

7.5 Layout and Spacing

As mentioned in section 4, crowding affects perception the most. It seems advisable to prefer wider spacing between elements.

Additionally, the number of components per side should be limited to three at most, anchoring them spatially to the corners and sides of the display and keeping the center of vision free.

7.6 Attention Management

Critically examine how information should be handled. Does the user need to be interrupted and does he have to do sth about it?

Dont overdo it with attention-grabbing features, the more contrast to other elements, the more they pop. If everything's the same risk inattentional blindness

The billboard style occludes much of reality around the text, which should generally be avoided. However, exactly this feature makes it suitable for high priority notifications.

7.7 Technical

8 Conceptualization

8.1 Relevant Parameters

A prioritization of elements is necessary for decisions on visual hierarchy, as well as which ones can be left out if there is not enough space.

	Display Mode	Visualization
Oxygen	Continuous	Status Bar, Value
Hazard Proximity	Pop-Up	Icon
Warning Notification	Pop-Up	Billboard
Working Time	Pop-Up, Continuous	Countdown
Heart Rate	Continuous	Icon, Value, Trend
Ambient Temperature	Continuous	Icon, Value, Trend
Humidity	Continuous	Icon, Value
Checklist	Continuous	Line Highlight
Tool Help	Pop-Up	Icon, Value

Table 1: Parameters chosen for visualization, ordered from most to least important.

Continuous means that the component will be displayed permanently. Parameters marked with pop-up will be displayed occasionally when relevant.

8.2 Components

8.2.1 Visualizing Oxygen Usage Through a Status Bar

Since the oxygen value is the crucial factor whether or not to continue work, this is the component taking the spotlight. As the most critical parameter, this component will be visible in any version of the HUD. To check the current status as quickly as possible, the value will be visualized with a status bar.

Since mechanical gauges with a pointer show visual progress by default, a radial version of the component was included. However, as mentioned in section 6, linear design enable much quicker parsing of the information. Compared to the linear version, it also takes up much more space to make the information readable.

The second version places the value below the bar, on the right. When the amount of oxygen decreases, the empty portions of the bar are only a contour. Thus, the contrast between value and bar becomes stronger. The label is placed

Figure 8: The initial concepts for the status bar, considering overall shape.

in front of the bar so they blend together, making it less salient. Additionally, the status bar was divided into quarter segments, enabling more accurate estimation of the amount left, without looking at the value.

Figure 9: The modified status bar and the following segmentation.

Depending on available screen space and layout configuration, a vertical version of the component might be needed in addition to the horizontal one.

Grouping the label and value on one side saves space (left and middle of figure 10). Aligning them at the top and bottom respectively accounts for visual crowding, making the value more distinctive. The oxygen value is located so that the status bar is seen first, i.e. if the component is placed on the left side, the bar will be on the right to maintain an optical guide.

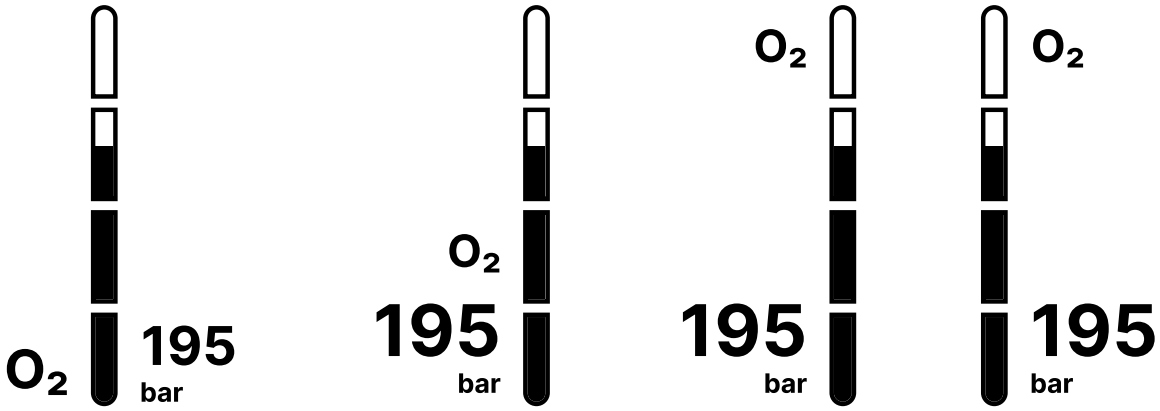

Figure 10: Development of the vertical status bar.

8.2.2 The Base Component

This element is built with a simple configuration for secondary information that is helpful, but does not need to be prioritized. This includes data like heart rate, but also environmental information such as ambient temperature or humidity.

Values that are not to be visualized are similar enough to be based on a universal base component. Apart from the actual parameter value, any of these elements can be hidden. This allows flexible configuration depending on the type of information.

Figure 11: The base component with configurable icon, parameter value, unit and trend indicator.

The icons replace labels, and should therefore be as unambiguous as possible. Leaving the unit out saves space, though in some cases, it supports the identification of the value (e.g., degrees Fahrenheit or Celsius).

The trend indicators are displayed based on the difference of the current and previous value. The value increasing or decreasing an unusual amount might indicate events to take notice of. For dynamic elements like these, the frequency of changes should not be too high to avoid distraction. Comparing the short and long arrows, the long arrows are better readable when out of focus.

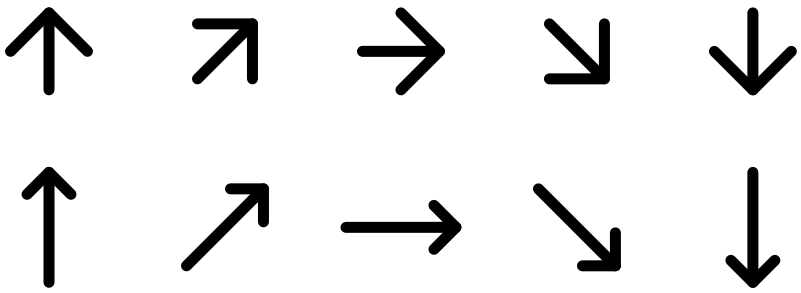

Figure 12: Comparison of short and long arrows.

All of these values are displayed continuously, with the exception of the time

component. It will only be displayed after a configured amount of time. The element popping up is meant to be perceivable in the peripheral vision, marking (for example) the halfway point, encoding this action with information. Since working time is limited by various parameters and therefore not fixed, the component is formatted as a countdown. To avoid pulling unnecessary attention, the display is limited to minutes, rather than ticking down by the second.

8.2.3 Notifications and Pop-Ups

Referncing typical industrial warning signs, the hazard icons were created in a triangular and a diamond-shaped version. Comparing them however, reveals that for the same size, the diamond shape appears smaller. This also means that there is less space for the graphic inside it. Avoiding unique silhouettes, units them in their function. Their placement is therefore more important for peripheral distinction. The icons are placed close together so that they are visible at the same angle if multiple hazards are present. Similar to a car's dashboard warning lights, each of them has a fixed position, facilitating rapid information acquisition. With regular notifications—depending on which one appears first—the order of content would be different.

Figure 13: Example notification with billboard style.

The core example of the notification system is the high priority warning. This component is designed with a so-called billboard style, meaning text with a surrounding box. The big, bright area ensures high visibility even if the pop-up is not in the center of vision. Since it is the highest priority, it is the only element placed in the upper central screen area. To avoid blocking central vision for too long, the pop-up will transform into a smaller version located at [...].

8.2.4 The Checklist Component

Since this element is text-based, it relies less on peripheral vision. To anchor the gaze to the current task, it is centered vertically and more distinctive than the other

displayed tasks.

Note that this component is meant as a visual example, which is why certain aspects of functionality might not work in regard to handling the list. (e.g. if only two next tasks are displayed, there is no way to see if new tasks got added → task number display in dynamic situations, generally better for protocol).

The icon to the right is an anchor for possible gaze-based interaction, in this case to show more details. This design keeps the current task centered in the detailed view as well for faster orientation. This also mitigates side effects from possible false positives—i.e. if the system mistakenly opens the detailed view.

8.3 UI Examples

Note: the following illustrations are not to scale.

The maximum variant of the interface was designed with full functionality in mind, based on Microsoft's HoloLens 2. It features a horizontal field of view of 43°, a resolution of 1440 by 936 pixels per eye, and provides full color availability (within the technical limits mentioned in section 3).

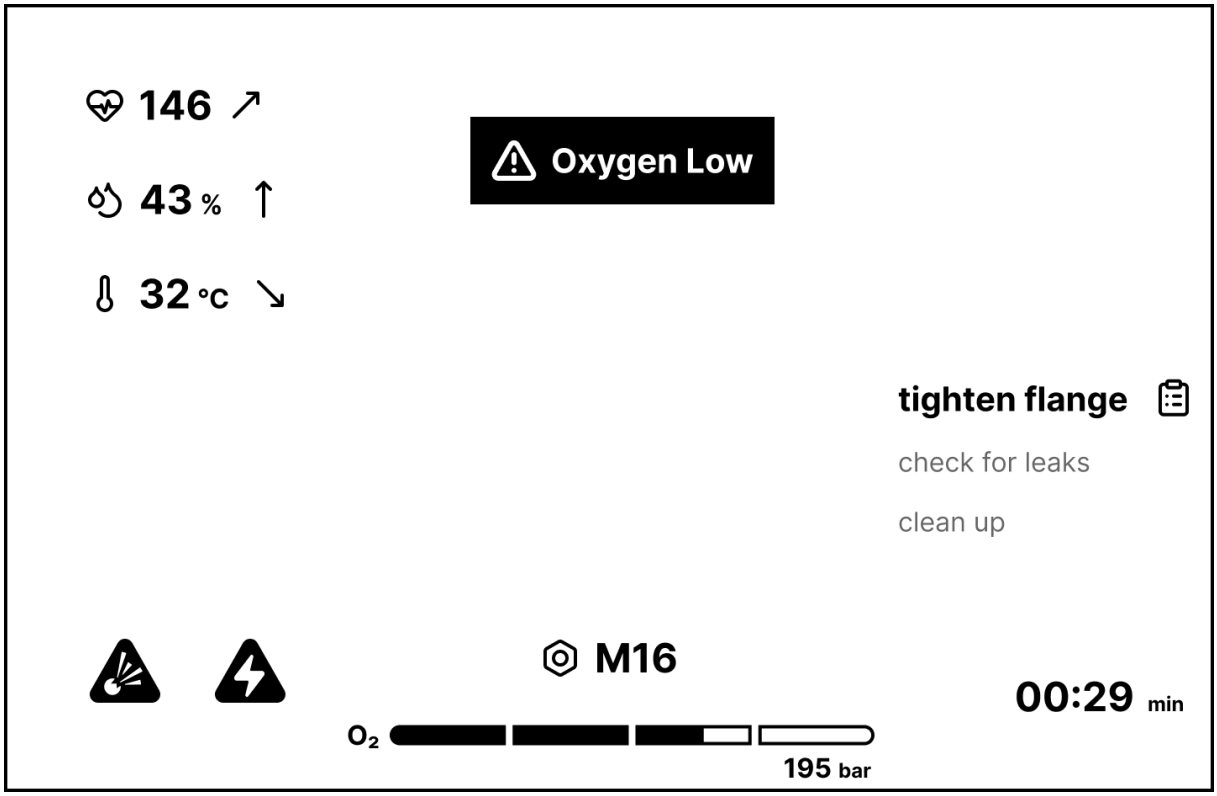

Figure 14: Full-featured UI concept for a binocular HUD.

While the oxygen component is the most important, it is not necessarily the one that is checked most often; however, it should be the easiest to check. Therefore, it is placed in the lower center. Occasional information pop-ups are placed above the oxygen bar, for the same reason: looking down is faster than looking up.

The minimum version is based on the Google Glass, featuring a monocular display with a resolution of 640 by 360 pixels.

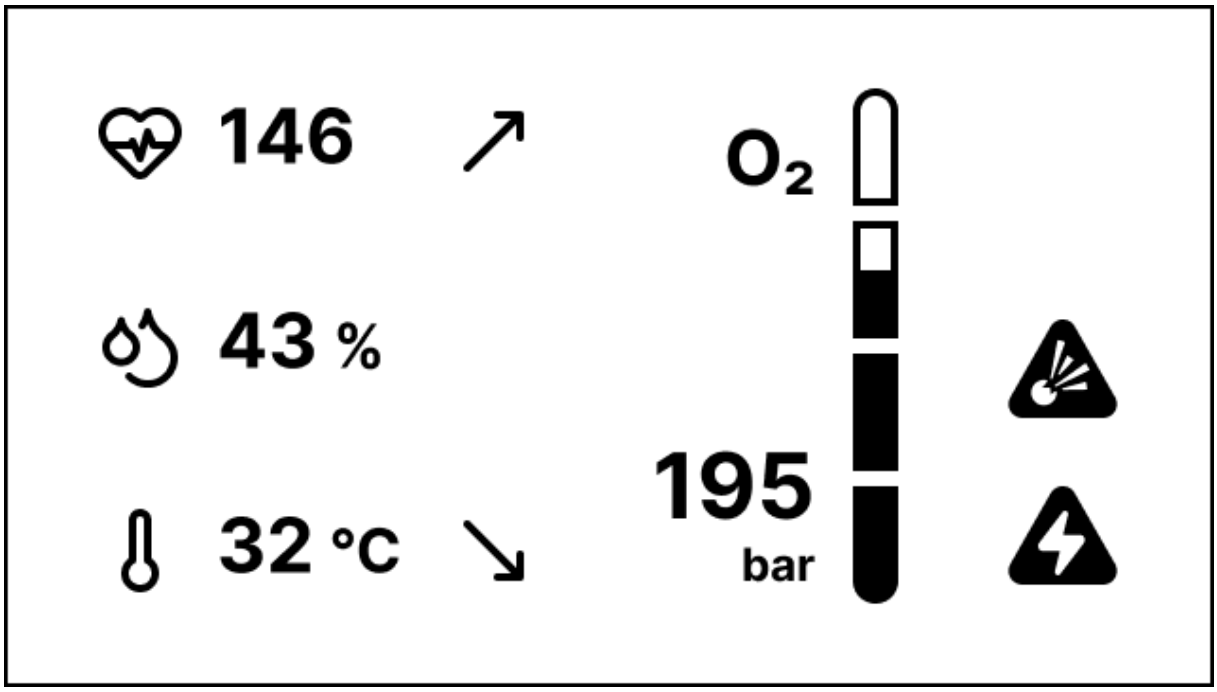

Figure 15: UI concept for a monocular HUD, worn in the upper left of the field of vision.

One could argue that some overlap of the components would be useful to make better use of the limited space. However, this would disrupt the grid pattern and the visual guides, resulting in longer search times.

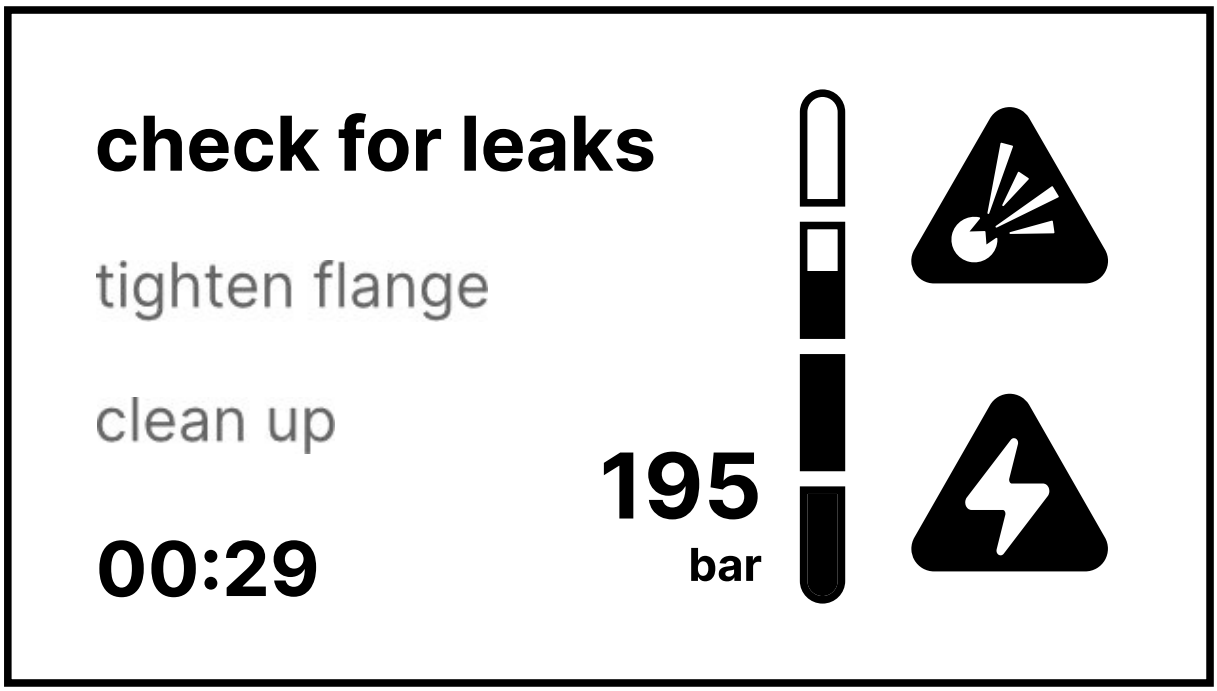

Figure 16: Overlap of the checklist and oxygen bar.

Figure 17: Size comparison of the Microsoft HoloLens 2 and the Google Glass Enterprise Edition 2.

9 Conclusion

Outlook

animation complete hands-free system attaching to control system multimodality: ZHU 2025

testing and iteration: Hari Chandana et al., 2023

Appendix

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